CODa Re-ID — Dataset Report

1. Authorship Information

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2. Research Areas .

Discipline: Computer Science

Field of study: Computer Vision, Robotics, Robot Autonomy

Theme of study: Object Re-Identification

3. Dataset Description

Mobile robots navigating real-world environments face a fundamental challenge: recognizing the same object across multiple encounters despite changes in lighting, weather, and viewing angle. CODa Re-Identification (Re-ID) dataset addresses this critical problem with over one million observations of hundreds of static outdoor objects captured on a university campus across different times and weather conditions.

We built CODa Re-ID dataset from the UT Campus Object Dataset (CODa).

Zhang, Arthur; Eranki, Chaitanya; Zhang, Christina; Hong, Raymond; Kalyani, Pranav; Kalyanaraman, Lochana; Gamare, Arsh; Bagad, Arnav; Esteva, Maria; Biswas, Joydeep, 2023, "UT Campus Object Dataset (CODa)", https://doi.org/10.18738/T8/BBOQMV, Texas Data Repository, V2

From the 53 categories of objects identified in CODA, we re-identified 8 categories: trees, poles, bollard, informational signs, traffic signs, fire hydrants, emergency phone, and trash can.

The pipeline followed is straightforward:

1. Align multiple robot trajectories.
2. Annotate instance-level 3D bounding boxes for every unique object.
3. Project those boxes into 2D images and refine the mask with segmentation.

Each object has a globally unique identifier, allowing it to be tracked consistently from sunny afternoons to rainy nights ,and from close-range to long-range views.

CODa Re-ID enables researchers to develop and benchmark object re-identification algorithms that work in real-world conditions. This dataset is essential for advancing object-based SLAM, long-term robot navigation, and persistent environmental understanding. These capabilities are required for robots that must operate reliably as their environments change over time.

4. Dataset Contents

- 2D annotations – object annotations with instance labels, 2d bounding boxes, segmentations mask – 1,037,814 annotations
- 3D bounding boxes – global instance annotations in world frame for each unique object – 557 object instances

5. Research Method

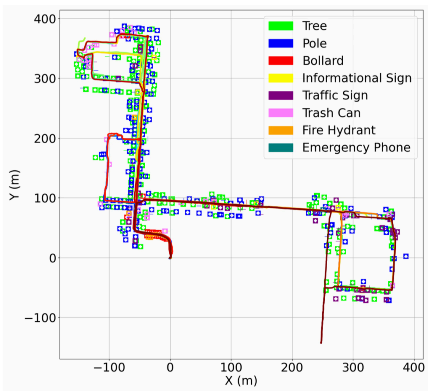
Research method type/s:

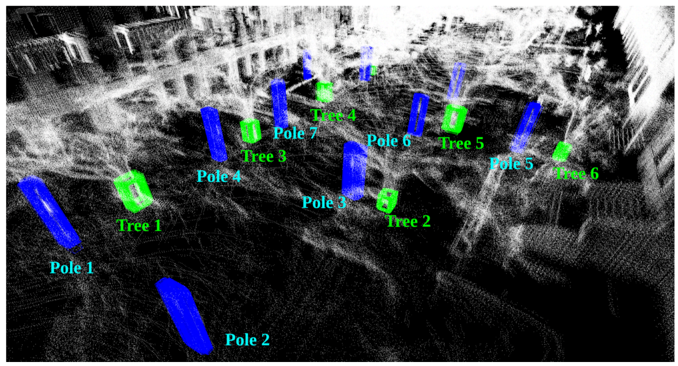
- Machine Learning

Data Generation Processing:

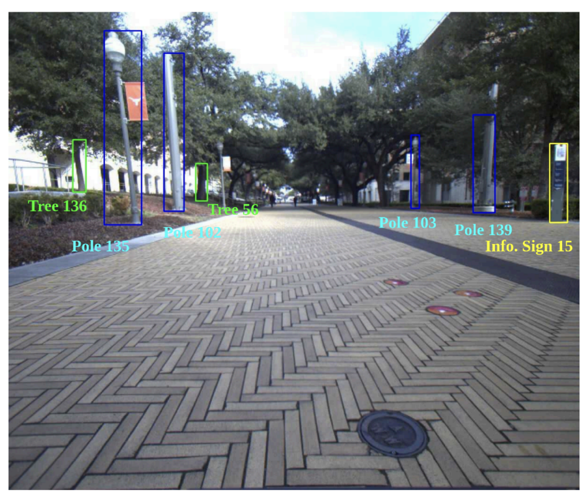
We created the CODa Re-ID dataset by post-processing the existing UT Campus Object Dataset (CODa). Our processing pipeline consisted of four steps:

1. Global Trajectory Alignment: We aligned all robot trajectories from the original dataset using interactive SLAM with manual loop closures to ensure global spatial consistency.
2. 3D Instance Annotation: We transformed per-frame 3D bounding boxes to the global coordinate frame using aligned poses, then clustered them to identify unique object instances. Each cluster received a globally unique instance ID, with representative 3D bounding boxes computed from the median of clustered annotations.





3. 2D Projection: We projected the global 3D instance annotations onto 2D images by inscribing ellipsoids in the 3D boxes and projecting them to image coordinates, adding configurable margins to preserve contextual information.
4. Segmentation and Refinement: We generated segmentation masks using The Segment Anything Model (SAM) and refined the 2D bounding boxes to better fit the mask contours, addressing projection misalignments.



- ***Data Dictionary:***
    - Instance ID: globally unique instance id
    - Class Label: semantic category
    - Sequence ID: trajectory sequence id
    - Weather Condition: environmental label based on sequence condition
    - bbox_2d: refined 2d bounding box [x1, y1, x2, y2]
    - bbox_3d: 3d bounding box [cx, cy, cz, l, h, w, r, p, y] of globally unique instance
    - Camera Pose: camera pose in world frame

6. Dataset:

Dataset Organization:

```
├── 3d_bbox/
    ├── global/
        ├── {object}.json
├── annotations/ (the tree structure is contained in the annotation.tar.gz )
    ├── {cam}
        ├── {sequence}
            ├── {frame}.json (each {frame}.json contains information about instance exists within the frame)
```

```
{
  "info": {
    "image_file": "2d_raw_cam0_0_3000",
    "img_size": [1224, 1024],
    "camera": "cam0",
    "sequence": 0,
    "frame": 3000,
    "weather_condition": "cloudy",
    "pose": [-49.782567, 160.291186, -0.989485, -0.666335, 0.744039, -0.027613, 0.040514]
  },
  "instances": [{
    "class": "Pole",
    "id": 122,
    "bbox": [747, 0, 844, 475],
    "segmentation": [
      [747, 0, 748, 64, 814, 68, 823, 19, 806, 474, 825, 456, 843, 5]
    ]
  }, {
    "class": "Pole",
    "id": 66,
    "bbox": [647, 237, 669, 446],
    "segmentation": [
      [660, 237, 647, 254, 653, 280, 649, 445, 656, 442, 659, 286, 668, 257]
    ]
  }, {
    "class": "Pole",
    "id": 69,
    "bbox": [1061, 178, 1099, 371],
    "segmentation": [
      [1092, 178, 1084, 179, 1075, 193, 1079, 223, 1061, 359, 1067, 370, 1075, 351, 1087, 222, 1098, 197]
    ]
  }, {
    "class": "Tree",
    "id": 51,
    "bbox": [246, 370, 274, 436],
    "segmentation": [
      [250, 372, 246, 381, 246, 406, 250, 430, 253, 433, 269, 435, 273, 433, 266, 372, 265, 370]
    ]
  }]
}
```


7. Segmentation / Annotation

To find information about the segmentation in the original CODA dataset look here: https://dataverse.tdl.org/dataset.xhtml?persistentId=doi:10.18738/T8/BBOQMV)

The instance-level annotations were generated through a systematic pipeline combining automated processing with manual verification.

3D Instance Generation: Per-frame 3D bounding boxes from the original CODa dataset were transformed to a global coordinate frame using aligned trajectories, then clustered with DBSCAN to identify unique object instances. Each cluster received a globally unique ID (0-556), with representative 3D boxes computed from median values.

2D Projection: Global 3D annotations were projected onto images by inscribing ellipsoids and projecting to 2D coordinates. Configurable margins were added to preserve contextual information around objects.

Segmentation Masks: The Segment Anything Model (SAM2) generated initial masks using projected bounding boxes as prompts. These were refined to improve boundaries and handle occlusions, with 2D boxes adjusted to fit final mask contours.

Quality Control: Manually checked the subset

8. Machine Learning

We trained CLOVER, a supervised contrastive learning approach that produces robust object embeddings invariant to viewpoint and lighting. CLOVER builds on an adapted DINOv2 (ViT-L/14) backbone augmented with lightweight, learnable adapter blocks. During training we freeze the ViT backbone and optimize only the adapters, using a supervised contrastive learning. Unlike methods that require foreground segmentation, CLOVER explicitly retains surrounding context and uses it as a discriminative signal for identification.

Code & resources

- GitHub: https://github.com/ut-amrl/clover
- Paper (arXiv): https://arxiv.org/abs/2407.09718

Performance Model Metrics

We evaluate re-identification using 1) mAP and 2) top-k recall for objects seen during training and for unseen objects

Object Instances	mAP	Top-1	Top-5
Seen during training	0.811	0.876	0.939
Unseen during training	0.519	0.800	0.882

Additional experimental results can be found in the paper.

9. Quality Control Statement

- **Scope.** Our goal was process verification,. We performed random spot-checks to confirm the end-to-end pipeline (trajectory alignment → 3D boxes → 2D projections → segmentations) behaves as intended.
- **Sampling.** We randomly sampled *1000* objects and *50* image instances (≈ *0.1*% of the dataset) and manually inspected: (i) projection correctness, (ii) ID consistency across views/conditions, and (iii) obvious annotation failures (e.g., empty/shifted boxes, mask leakage).
- **Limitations.** No dataset-wide de-duplication, outlier removal, or blanket relabeling was performed. Residual errors may remain outside the sampled subset.

10. License

This dataset is distributed under the same license of the original CODA dataset.

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